

From budgets to barriers and backlash: learning from New Zealand's difficulties implementing limits on reactive nitrogen

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KEY CLAIM

Progressing from assessment budgets to managing N excess requires footprints and efficiency metrics to function as a system for environmental integrity.

BRIEF JUSTIFICATION, REMAINING CHALLENGES

Aotearoa New Zealand had many advantages that would seem to have allowed it to progress smoothly from N budgets to footprints and efficiency metrics. Key examples include:

- National and regional N budget methodology, including monitoring, historical assessment and scenarios that generated policy action (Parfitt et al 2006, 2008, 2012).
- The historic dominance of nitrogen fixation in pastures presents a self regulating alternative to industrial fertilisers.
- A reputation for leadership in agricultural footprinting, greenhouse gas inventory and measurement.
- A history of innovative policy design including the first comprehensive national emissions trading scheme (ETS) and scheme for N cap and trade in a catchment (OECD, 2015).
- A sense that the national brand generates value by transmitting a 'clean, green' image representing both health and environment.

Yet, recent years have seen more backlash than implementation, with hard-won technocratic and policy gains struggling or being rolled back nationally and regionally.

- The model system used by farms to assess emissions to water was found not fit for many regulatory roles it had evolved to fill, creating a loss of confidence and need for years of rebuilding and refocussing to greenhouse gases (PCE 2018, MPI 2021).
- Although political resolve existed to address nitrogen as the key source of eutrophication in freshwater following the 2017 election, close elections and the dominance of agriculture in export earnings has made implementation of limit setting difficult, leading to rollback or repeal of legislation and policy setting effective environmental limits.
- Commitments to begin to include agriculture in the comprehensive ETS were abandoned.
- As a result, efforts to quantify and focus on the main period of N excess in pastoral systems have neither become visible nor incentivised for reductions.

New Zealand's case presents a cautionary tale, from which I suggest hope lies in successful implementation of a system focussed more on linking budgets, footprints and efficiency metrics than in ideal but disconnected metrics. Such a system benefits from being designed for high environmental integrity – consistency across science, policy and legislation, as well as transparency and simplicity so that assessment and monitoring of progress is accessible not just to governments and corporations along supply chains, but also NGOs, communities and consumers. This places nitrogen management in a context where it can and should be considered along with climate change and biodiversity initiatives that will also benefit from the consideration of principles for environmental integrity.

POTENTIAL FOR SYNERGY WITH OTHER CONTRIBUTIONS

The purpose of this submission is to propose that discussion focus on linking across budgets, footprints and efficiency metrics to achieve system integrity for environmental management, including but not limited to nitrogen. An emerging filing questioning New Zealand's adherence to environmental and climate clauses in the New Zealand-European Union Free Trade Agreement may serve as a point nitrogen metrics and their integrity can be applied across scales.

QUESTIONS TO SEED THE DISCUSSION

1. How can N budgets, footprints and efficiency metrics provide transparent scenarios driving political economies and supply chains to more effectively manage N use and excess?
2. What connections between the scales of N budgets, footprints and efficiency are most effective at enabling innovation while enhancing the resilience and integrity of the overall system by retaining transparency while resisting politically or economically driven instability or backlash?

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